# RAG-Based Clinical Decision Support / Triage Assistant

## Complete Capstone Project Document

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## 1. Literature Review (2020–2026)

**Foundational RAG (2020).** Lewis et al. (2020, NeurIPS) introduced retrieval-augmented generation for knowledge-intensive NLP tasks — the base architecture (retriever + generator) nearly all later clinical-RAG work extends.

**Hallucination as the core motivating problem (2023–2024).** Clinical-LLM literature is unusually explicit about the stakes here: <cite index="26-1">prior studies reported hallucination rates of 28.6–39.6% in medical-specialized models</cite> without retrieval grounding, and a JAMA Network Open study specifically examined accuracy/consistency/hallucination when LLMs analyze unstructured clinical notes. This is the direct justification for why RAG (not a bare LLM) is the right architecture for clinical decision support.

**Domain-grounded RAG systems (2024).** Multiple guideline-grounded systems emerged across specialties: a hepatology-guideline RAG framework (Kresevic et al., *npj Digital Medicine*, 2024), and later ophthalmic and radiology-specific variants. These consistently show the same pattern — narrow, specialty-specific grounding reduces hallucination substantially but each system is built and evaluated in isolation with inconsistent methodology.

**Architecture sophistication (2024–2025).** Recent variants extend beyond plain dense retrieval: <cite index="24-1">hybrid RAG combining dense and sparse retrieval with reciprocal rank fusion, and knowledge-graph-enhanced retrieval over UMLS (a ~3M-node clinical concept graph) with graph-attention-network-based relation scoring</cite> represent the current state of the art in retrieval sophistication for this domain. Privacy-preserving, on-premises deployment variants (e.g., "MediRAG") specifically address the fact that clinical data often cannot leave institutional infrastructure — <cite index="24-1">though this vital privacy protection introduces additional system complexity.</cite>

**Quantified hallucination-reduction results (2025).** A radiology contrast-media consultation study found <cite index="26-1">complete elimination of hallucinations (reducing incidence from 8% to 0%) when using a RAG-enhanced local LLM</cite> compared to the ungrounded baseline — a strong, citable, concrete result for your own evaluation framing.

**RAG vs. fine-tuning debate (2025).** A systematic review (Jan 2023–Dec 2025 search window) concluded RAG is generally the more resource-efficient and scalable approach for medical LLM deployment, since fine-tuning requires substantial compute and annotated data often unavailable in clinical settings — though several studies found hybrid RAG+fine-tuning approaches enhance accuracy further in specialized diagnostic tasks.

**Evaluation methodology remains the weak point (2024–2026).** This is your most important gap: work on "Ranking Over Scoring" (COLING 2025) argues for more reliable, robust automated evaluation of LLM-generated medical explanations, and a broader trustworthy-RAG survey (2025) catalogs LLM-as-judge as an emerging but still-inconsistent evaluation practice. Across nearly every paper reviewed, hallucination-rate measurement methodology differs (human review vs. automated fact-checking vs. LLM-as-judge), making cross-paper comparison unreliable.

**Summary table**

| Period | Focus | Key contribution |
|---|---|---|
| 2020 | Foundational architecture | RAG introduced for knowledge-intensive NLP (Lewis et al.) |
| 2023–2024 | Problem motivation | Hallucination rates of 28.6–39.6% documented in ungrounded medical LLMs |
| 2024 | Domain-grounded systems | Specialty-specific guideline-grounded RAG (hepatology, ophthalmology) |
| 2024–2025 | Architecture sophistication | Hybrid dense+sparse retrieval; knowledge-graph-enhanced (UMLS) retrieval |
| 2025 | Quantified results | RAG reduces hallucination from 8% to 0% in a radiology-consultation study |
| 2025 | RAG vs fine-tuning | RAG favored for resource-constrained clinical deployment; hybrid approaches gaining traction |
| 2024–2026 | **Evaluation gap** | No standardized hallucination-measurement methodology across studies — your opening |

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## 2. Research Gap Analysis

1. **No standardized, reproducible hallucination-evaluation protocol.** Every paper measures hallucination differently (human review, automated fact-checking, LLM-as-judge), making results incomparable — this is explicitly flagged across multiple 2024–2026 sources above.
2. **Confidence-calibrated abstention is underexplored.** Most systems always answer; very few explicitly detect low-retrieval-confidence situations and abstain or flag uncertainty — a real patient-safety gap, since a confidently wrong answer is more dangerous than a system that says "I'm not sure, consult a specialist."

3. **Narrow, single-specialty focus limits generalizable conclusions.** Existing systems (hepatology, ophthalmology, radiology) are each built and evaluated in isolation — a cross-domain evaluation using one consistent methodology hasn't been done at the capstone-accessible level.

**Your novelty claim should combine gaps #1 and #2:** a **guideline-grounded RAG assistant with confidence-calibrated abstention**, evaluated using a rigorous, reproducible hallucination-measurement protocol (combining automated fact-checking against retrieved source passages + structured human review) — directly addressing the methodology-inconsistency gap while adding a genuine safety mechanism (abstention) that most reviewed systems lack.

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## 3. Existing Solutions Comparison

| System/Approach | Strength | Weakness (your opening) |
|---|---|---|
| Vanilla LLM (no retrieval) | Simple, fast | 28.6–39.6% hallucination rate in medical-specialized models — unsafe for clinical use |
| Hepatology/Ophthalmology guideline-RAG (2024) | Strong specialty-specific hallucination reduction | Narrow scope, isolated evaluation methodology, no abstention mechanism |
| Hybrid dense+sparse + UMLS knowledge-graph RAG | State-of-the-art retrieval sophistication | Complex to build in 3–4 months; doesn't address the evaluation-standardization or abstention gaps |
| MediRAG (privacy-preserving, on-premises) | Addresses real data-residency constraint | Added complexity; privacy focus, not evaluation-rigor or abstention focus |
| Fine-tuned medical LLM (no RAG) | Can outperform RAG on narrow diagnostic tasks in hybrid setups | Requires substantial compute/annotated data — impractical for a 3–4 month capstone |

**Recommendation:** Build a **dense-retrieval RAG pipeline** (skip the full knowledge-graph complexity — it's not your novelty axis and would eat your timeline) using a vector DB (e.g., pgvector or Chroma) over a curated guideline corpus, and invest your saved time into the **abstention mechanism and evaluation protocol**, which is where your actual contribution lives.

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## 4. Novel Contribution (your paper's thesis)

> **"Confidence-Calibrated Retrieval-Augmented Generation for Safe Clinical Decision Support: A Reproducible Hallucination-Evaluation Protocol"**

A guideline-grounded RAG system that (a) explicitly scores retrieval confidence per query and abstains/flags uncertainty below a threshold rather than always answering, and (b) is evaluated using a documented, reproducible protocol combining automated citation-grounding checks (does the answer's claims trace back to retrieved passages) with structured human review — directly addressing the standardization gap the literature repeatedly flags, while adding a safety mechanism (abstention) most existing systems don't have.

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## 5. Patentability Analysis

**Low patentability — this should be a pure publication play, and that's fine.**

- RAG architecture itself, and even confidence-thresholded abstention, are not novel *mechanisms* in the abstract (confidence thresholding exists broadly in ML) — the novelty here is in application and evaluation rigor, which is publishable but not independently patent-eligible.
- **No realistic patent path here** — don't spend capstone time on IP filing for this one. Focus entirely on making the evaluation methodology rigorous enough to be genuinely citable; that's your real asset.

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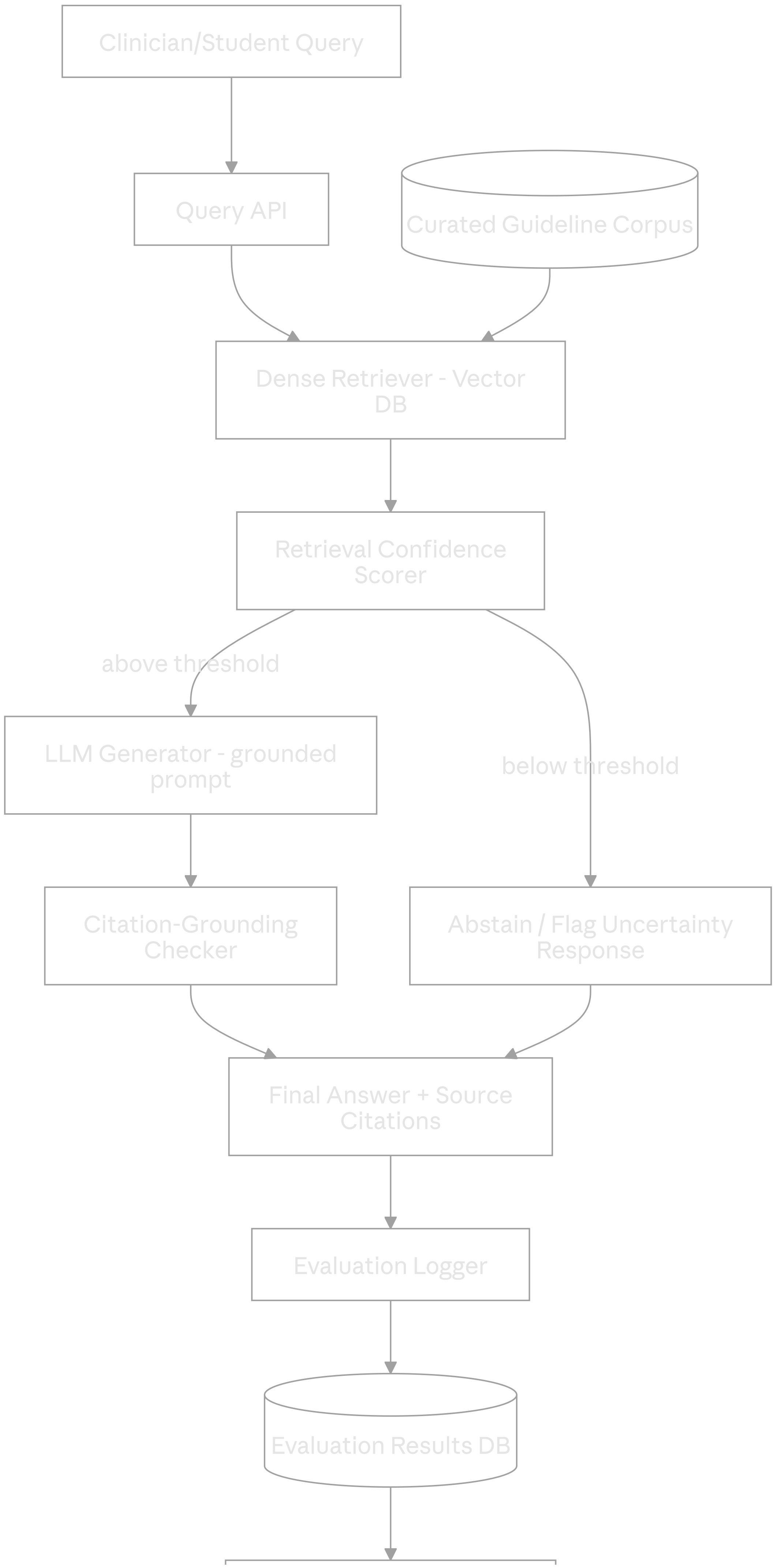
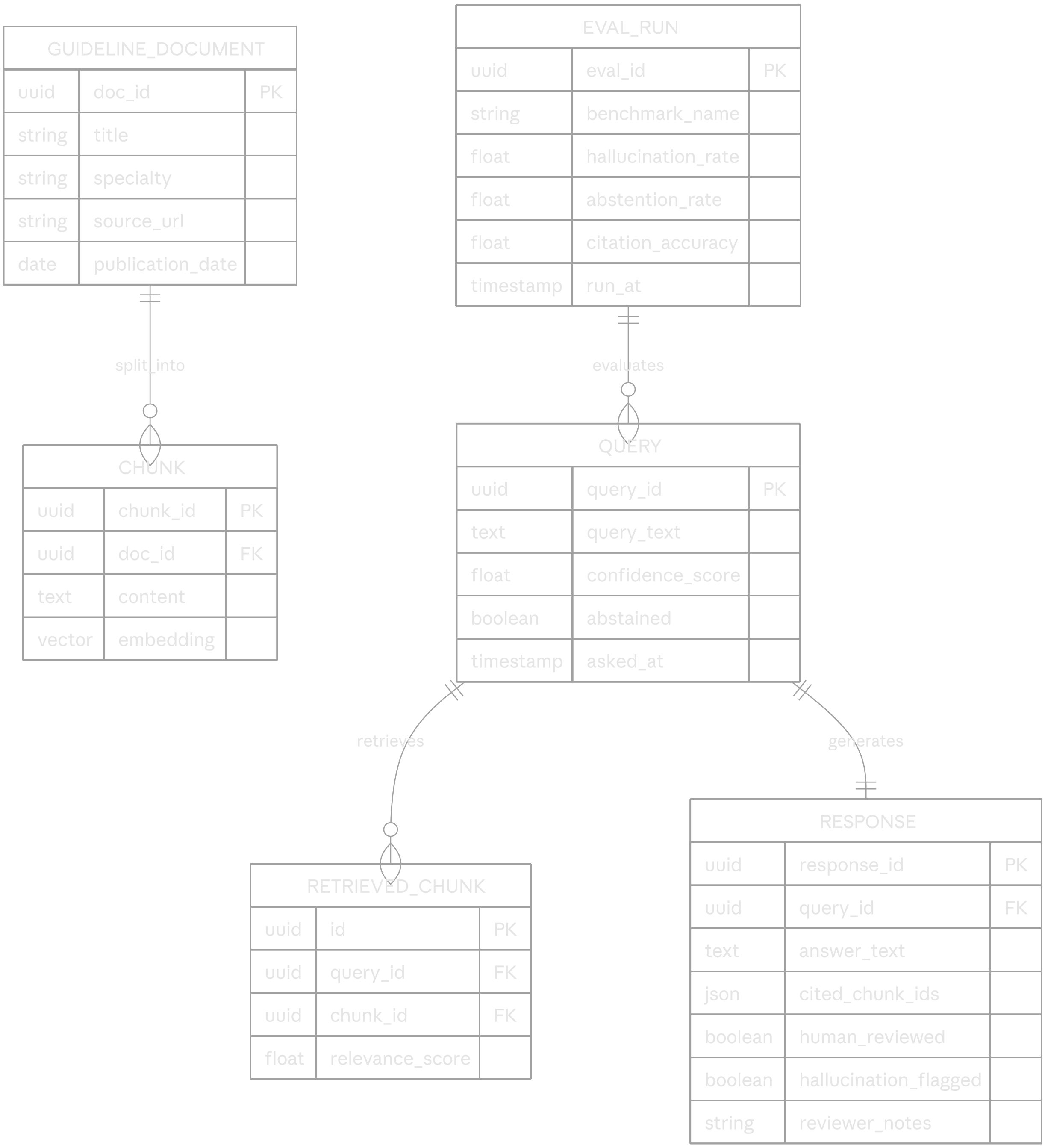
# 6. Complete System Architecture



Clinician/Student Query
Query API
Curated Guideline Corpus
Dense Retriever - Vector DB
Retrieval Confidence Scorer
above threshold
below threshold
LLM Generator - grounded prompt
Citation-Grounding Checker
Abstain / Flag Uncertainty Response
Final Answer + Source Citations
Evaluation Logger
Evaluation Results DB


## 7. Database Schema (ER Diagram)



## 8. API Design

| Method | Endpoint | Purpose |
| --- | --- | --- |
| POST | `/auth/login` | User authentication (JWT) |
| POST | `/guidelines/upload` | Ingest a new guideline document into the corpus |
| POST | `/query` | Submit a clinical question, get grounded answer + citations or abstention |
| GET | `/query/{id}` | Retrieve a past query's full response + retrieved chunks |
| POST | `/evaluation/run` | Trigger an evaluation run against a benchmark (e.g., MedQA subset) |
| GET | `/evaluation/{id}/results` | Fetch hallucination-rate, abstention-rate, citation-accuracy results |
| POST | `/review/{response_id}` | Submit human-review annotation (hallucination flag, notes) |
| GET | `/corpus/stats` | Corpus size, specialty coverage, chunk count |

## 9. Frontend Wireframes

```
┌─────────────────────────────────────────────────────────┐
│  Clinical RAG Assistant            [Login: clinician]   │
├─────────────────────────────────────────────────────────┤
│  Ask a question:                                        │
│   ┌─────────────────────────────────────────────────┐   │
│   │ "What's the recommended first-line treatment for..." │   │
│   └─────────────────────────────────────────────────┘   │
│  [Submit]                                               │
├─────────────────────────────────────────────────────────┤
│  Answer (confidence: 0.87 — high)                       │
│  "Based on [Guideline X, Section 3.2]..."               │
│  Sources: [1] Guideline X  [2] Guideline Y              │
│                                                         │
│  --- OR, low-confidence case ---                        │
│  ⚠ Confidence: 0.31 — Insufficient grounding found.     │
│  This question may fall outside the current guideline   │
│  corpus. Please consult a specialist.                   │
├─────────────────────────────────────────────────────────┤
│  [Researcher view] Evaluation Dashboard                 │
│  Hallucination rate: 4.2%   Abstention rate: 11%        │
│  Citation accuracy: 93%                                 │
└─────────────────────────────────────────────────────────┘
```

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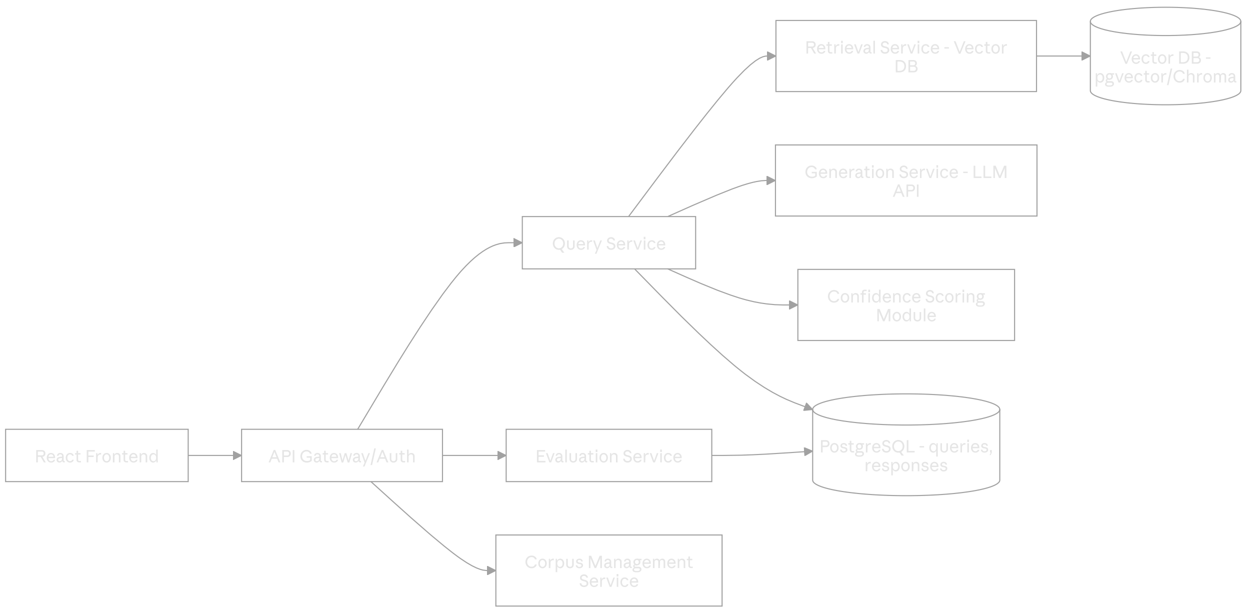
## 10. Backend Microservice Architecture




Modular monolith is again the right call here for 2 builders — separate Query, Retrieval, Confidence, and Evaluation as internal modules/services within one deployable app rather than true separate microservices.

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## 11. AI/ML Model Design

- **Retriever:** dense embedding retrieval (e.g., a strong open embedding model) over chunked guideline text, stored in pgvector or Chroma.
- **Generator:** an LLM API call with retrieved chunks injected into the prompt, explicitly instructed to cite which chunk(s) support each claim (this citation requirement is what enables your citation-grounding check).
- **Confidence scorer:** combine (a) top-k retrieval similarity scores and (b) an LLM-as-judge or entailment-model check of whether the generated answer's claims are actually supported by the retrieved chunks — abstain if either signal is below threshold.
- **Citation-grounding checker:** for each claim in the answer, verify it can be traced to a specific retrieved chunk (simple entailment check or exact/paraphrase match against chunk content) — this is your automated half of the hallucination-evaluation protocol.
- **Threshold tuning:** treat the abstention threshold as an experimental variable — report the precision/coverage tradeoff (higher threshold → fewer hallucinations but more abstentions) as a curve, not a single fixed number.

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## 12. Dataset Recommendations

| Dataset/Corpus | Use | Why it fits |
|---|---|---|
| **Curated public clinical guidelines** (e.g., WHO, NICE, or specialty-society guidelines, whichever are freely redistributable) | Your grounding corpus | Directly matches the "guideline-grounded" framing used in the 2024 literature you're extending |
| **MedQA / PubMedQA** | Evaluation benchmark | Standard, widely used medical QA benchmarks — enables direct comparison to prior RAG-vs-vanilla-LLM results |
| **MedHALT** (Med-Halt hallucination test, referenced above) | Hallucination-specific evaluation | Purpose-built for measuring medical hallucination — strengthens your evaluation-rigor claim |
| **A held-out "out-of-corpus" question set** (questions outside your guideline corpus's scope) | Abstention evaluation | Essential for measuring whether your system correctly abstains rather than hallucinating on unfamiliar topics |

**Scope note:** pick ONE clinical sub-domain (e.g., primary-care triage, or a single specialty) rather than trying to cover all of medicine — this keeps your guideline corpus curatable in the available time and matches how the existing literature is (necessarily) specialty-scoped.

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## 13. Evaluation Metrics

- **Hallucination rate:** % of answered (non-abstained) responses containing at least one unsupported claim — measured via your automated citation-grounding checker, spot-validated by human review
- **Abstention rate:** % of queries where the system declines to answer — report alongside hallucination rate as a tradeoff curve, not in isolation
- **Citation accuracy:** % of cited sources that genuinely support the claim they're attached to
- **Coverage-accuracy tradeoff:** as you raise the confidence threshold, hallucination should drop but abstention should rise — plot this curve, it's your key result
- **Comparison baselines:** vanilla LLM (no retrieval), RAG without abstention (always answers), your full system — report hallucination rate for all three
- **Human-agreement check:** inter-rater agreement between your automated citation-checker and human reviewers on a sample — validates your automated metric is trustworthy, addressing the evaluation-standardization gap directly

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## 14. Week-by-Week Implementation Plan (16 weeks)

| Week | Milestone |
|---|---|
| 1 | Literature review finalized; pick clinical sub-domain; team roles locked |
| 2 | Guideline corpus curated and chunked; vector DB set up |
| 3 | Basic dense retrieval pipeline working (query → top-k chunks) |
| 4 | LLM generation with citation-instructed prompting integrated |
| 5 | Baseline vanilla-LLM (no retrieval) comparison system built for later contrast |
| 6 | Confidence scoring module v1 (retrieval-similarity based) |
| 7 | Citation-grounding checker v1 (automated claim-to-chunk verification) |
| 8 | Abstention mechanism wired end-to-end; internal review 1 prep |
| 9 | Internal Review 1 — present pipeline + preliminary hallucination-rate numbers |
| 10 | Confidence scorer v2: add LLM-as-judge or entailment-model signal, not just retrieval similarity |
| 11 | Evaluation harness built against MedQA/PubMedQA/MedHALT subsets |
| 12 | Full evaluation run: vanilla LLM vs RAG-no-abstain vs full system; threshold sweep for tradeoff curve |
| 13 | Human-review sample for inter-rater validation of automated metric; dashboard built |
| 14 | Internal Review 2 — present full results; incorporate feedback |
| 15 | Paper: Results/Discussion finalized; target-venue formatting |
| 16 | External Review 3; submission to target venue |

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## 15. Team Task Allocation (2 developers + 1 researcher)

### Developer A — Retrieval & Corpus

- Weeks 1–4: guideline corpus curation, chunking strategy, vector DB setup, retrieval pipeline
- Weeks 5–10: confidence scoring (retrieval-similarity component), corpus-management API
- Weeks 11–16: dashboard backend, evaluation-harness infrastructure, deployment/demo

### Developer B — Generation & Safety Mechanism

- Weeks 1–4: LLM generation integration, citation-instructed prompting
- Weeks 5–10: citation-grounding checker (this is core to your novelty — prioritize it), abstention logic, LLM-as-judge confidence signal
- Weeks 11–16: full evaluation-run execution, threshold-sweep analysis, results processing

**Researcher — Literature, Writing, Evaluation Design**

- Weeks 1–2: literature review, gap analysis, finalize novelty framing (evaluation-standardization + abstention)
- Weeks 3–10: draft Methods in parallel with build; design the human-review protocol (inter-rater agreement setup) *with* the developers so it's actually executable at week 13
- Weeks 11–16: Results, Discussion, Abstract, Introduction; venue formatting; review-panel prep; personally run/coordinate the human-review sample since this needs careful, unbiased protocol design

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## 16. IEEE / ACM / Springer Conference Recommendations

- **IEEE conferences** on Healthcare Informatics, Biomedical & Health Informatics (e.g., IEEE regional health-AI symposiums) — standard fit.
- **ACM CHIL-adjacent workshops** or ACM regional health-informatics symposiums — directly relevant to clinical NLP/RAG work.
- **Springer LNCS/LNAI volumes** on medical NLP or clinical AI — accessible for first-time student authors.
- **JMIR AI** and similar venues have published exactly this kind of clinical-RAG evaluation work recently (see literature review above) — worth checking their current submission scope even if primarily a journal rather than conference track, since a strong capstone result could be journal-appropriate too.

**Action item:** verify live CFPs/deadlines 2–3 months before your week-15 target.

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## 17. Resume Impact Analysis

**Resume bullet (example):**

> "Built a guideline-grounded RAG clinical assistant with confidence-calibrated abstention, reducing hallucination rate to X% (vs. Y% for ungrounded baseline) using a reproducible evaluation protocol combining automated citation-grounding and human review."

**Why this lands with recruiters in 2026:**

- RAG/LLM-grounding skills are currently the single hottest applied-AI competency in job postings — this project speaks directly to that demand.
- "Confidence-calibrated abstention" signals safety-conscious AI engineering, a differentiator as companies increasingly care about responsible-AI practices, not

just raw capability.

- A rigorous, documented evaluation methodology (not just "I built a chatbot") is exactly what separates a genuinely interview-worthy AI project from a weekend hackathon demo.
- Fast to build given your existing RAG experience, which means more of your 3–4 months goes into the evaluation rigor that actually makes this resume-differentiating.

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## 18. Interview Questions Based on This Project

1. Why RAG instead of fine-tuning for this use case — what's the actual tradeoff?
2. How does your confidence scorer decide when to abstain, and how did you pick the threshold?
3. Walk through your citation-grounding checker — how do you verify a claim is actually supported by a retrieved chunk, and where does it fail?
4. What's the difference between retrieval-similarity confidence and an LLM-as-judge confidence signal — why use both?
5. How did you validate that your automated hallucination metric is trustworthy (inter-rater agreement with human review)?
6. What happens when the guideline corpus itself is outdated or wrong — how would you handle that failure mode?
7. How would you extend this system to handle multi-turn clinical conversations rather than single-shot Q&A?
8. What are the real deployment blockers between this prototype and actual clinical use (regulatory approval, liability, integration with EHR systems)?

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## 19. Future Startup / Commercialization Opportunities

- **B2B health-AI infrastructure play:** license the confidence-calibrated RAG + evaluation framework to health-AI vendors or hospital systems building their own clinical assistants — the evaluation rigor itself (not just the chatbot) is a sellable asset, since the field's biggest documented weakness is inconsistent evaluation.
- **Realistic first customer:** medical education platforms or clinical-decision-support vendors who need a safer, better-evaluated alternative to a bare LLM wrapper — lower regulatory burden than direct patient-facing deployment.
- **Differentiator:** most current commercial "AI medical assistants" don't publicly report hallucination rates or have abstention mechanisms — genuinely leading with measured safety is a real trust/marketing wedge, not just a technical one.
- **Caution:** direct clinical deployment requires regulatory clearance (e.g., as a Software as a Medical Device in various jurisdictions) — realistic path is "medical-education or clinician-support tool with a multi-year path to regulated deployment," not an immediate consumer product.

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*All three projects (FL+DP, RL Microgrid, RAG Clinical) are now fully scoped at this depth. Want a final side-by-side comparison to help you commit to one, or should we start turning your top choice's week-1 tasks into an actual checklist?*